BT 404

M.C.S

Created by Amaan Khan

1. **(Phytochemicals)** that retard the development and progression of degenerative diseases.

2. The chemical components of food that which perform these functions are called **nutrients**

3. ___ study of various nutrients, their functions, food sources and their utilization by the human body. **Nutrition**

4. _____ is the application of food science & engineering principles to the selection, preservation, processing, packaging, distribution, and use of safe, nutritious. **Food technology**

5. In our diet, about **65 to 80** percent of the energy is supplied by carbohydrates.

6. Which one of the following the simplest carbohydrates? **Monosaccharide or Glucose**

7. Polysaccharides in nature consist of **hundreds to several thousand** of monosaccharide units.

8. Which one of the following included the nondigested carbohydrate? **Dietary fiber**

9. Oils and fats belong to a naturally occurring substances called _____ **Lipids**

10. _____ of Nitrogen are present in the protein. **16 %**

11. Proteins are built from simpler compounds called **amino acids.**

12. The number of minerals is required by human body for proper health and growth. **14**

13. _____ are essential for the maintenance of good health as they catalyse various body processes, which help to utilize all the nutrients supplied and regulate a number of functions in the body. **Vitamin**

14. Which one of the following is included water soluble Vitamin? **B and C Vitamin**

15. **Water Activity** is a concept different than water contents and it has important implications such as non-enzymic browning, lipid oxidization, degradation of vitamins, enzymatic hydrolysis, protein denaturation, and starch gelatinization and retro gradation.

16. Oligosaccharides consist of _____ units of Monosaccharides? **2 to 10**

17. The polysaccharide that animals use for the short term storage of food energy in the liver and muscles is known as **glycogen.**

18. Which one of the following carbohydrates is found most abundant in nature? **Cellulose**

19. **Hydrogenation** oils are exposed to hydrogen gas under conditions of high temperature (150– 180◦C) and pressure (2–10 atm) in the presence of a **nickel** catalyst.

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20. The human body consists of about **3%** minerals, most of which is in the skeletal system.

21. **Iron** is an essential part of both the blood pigment hemoglobin and the muscle pigment **myoglobin.**

22. _____ is associated with the function of musc1es and nerves and with the metabolism of carbohydrates. **Potassium**

23. If **6%** of the potassium contained in the cells were to escape into the blood the heart would stop.

24. About **85%** is found combined with calcium is _____ **Phosphorous**

25. Which Vitamin is important for vision and resistance to infection? **Vitamin A**

26. **Vitamin K** is essential for clotting of blood.

27. Biotin is reported to be a coenzyme in the synthesis of **aspartic acid**, involves fixation of **carbon dioxide**

28. Which is one of the brain's principal neurotransmitters? **Choline**

29. The name of the Vitamin C is _____ **Ascorbic Acid**

30. Pectin has **50 to 70 %** etherification and it sets slowly with addition of more sugar and acids.

31. Moisture contents **60-95 %** water contents in foods.

32. **Nitrogen** is a central element of proteins but it is also found in other compounds such as nucleic acids, alkaloids, nitrogen lipids and vitamins.

33. Carbohydrates are extracted from foods by boiling a defatted sample with an **80% alcohol** solution.

34. Sugars react with the anthrone reagent under acidic conditions to yield a **blue-green** color.

35. The **proteolytic** enzymes in the tenderization of meats, pectinases in the ripening of fruits, and trypsin inhibitors in legume seeds are proteins.

36. The term **‘‘dietary fiber’’** was originally synonymous with ‘‘**roughage**’’—the nondigestible plant matter, principally of cell wall origin.

37. Fats are built up by linking together a number of individual **fatty acids with glycerol**.

38. Each gram of pure fat or oil supplies **nine** kilocalories.

39. The specific gravity of oils and fats is about **0.9**

40. Thermal oxidation which is accelerated by the addition of **soda and water.**

41. One gram of fats and oils produces **9 kilocalories** of energy so they are concentrated form of energy.

42. **Bile** neutralizes acidic pH of food in duodenum so that the work of pancreas enzymes starts.

43. Lipids can form an emulsion with water when beaten up with it to form tiny globules in the presence of suitable emulsifying agent. **Emulsification**

44. The **smoke point** is the temperature at which a fat or oil gives off a thin **bluish** smoke.

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45. Spoilage of fats results in off flavour and renders the fat inedible. These changes are known as **rancidity** of fats.

46. **Hydrolysis** is the decomposition of fats (triglyceride molecules) to glycerol and free fatty acids.

47. **Oxidation** of unsaturated fats leads to oxidative rancidity.

48. UHT Milk owes its long life to ultra high temperature **132°C** for one second followed by packaging.

49. **Psychrotrophs** grow at **5°C** less and make up a large number of the microbes in milk.

50. Pasteurization provide a product with average shelf life **7-10 days**

51. Raw milk is associated with pathogenic diseases such as **tuberculosis, brucellosis or typhoid fever**

52. Holding Method - **62.7°C for 30 minutes**.

53. Higher Temperature Short Time Method (HTST) - **71.6°C at least for 15 seconds**.

54. Ultra High Temperature Process- Milk is heated to a temperature of **89–90°C or more for 1 second or less**.

55. Which of the following is the popular milk product in India? **Curd (Yogurt)**

56. When the pH reaches **4.6** the milk protein, casein, coagulates as curd.

57. The optimum temperature for the formation of curd is **35-40°C**.

58. Addition of lemon juice or citric acid to hot milk and precipitating the casein is called **Paneer**

59. Paneer contains about **18 percent** protein and is a good source of it.

60. The cheese formed is coated with **paraffin** to prevent loss of moisture and allowed to ripen at low temperature.

61. Cheese contains about **24 percent** protein and is thus a concentrated source of protein.

62. **Proteases** are enzymes that degrade proteins.

63. The amylase activity in milk is often referred as **diastatic power** and refers to the production of reducing substances by the action of alpha & beta amylases.

64. A malt infusion is added to the previously prepared limit dextrin substrate and aliquots removed periodically to a solution of **dilute iodine.**

65. The **iodine number** is a measure of degree of unsaturation, that is, number of carbon double bonds in relation to the amount of fat or oil.

66. The average protein content of meat ranges from **16 to 23** percent and the average fat content ranges from **10 to 40** percent.

67. In muscle tissue the smallest unit of a muscle is a **muscle fiber**

68. Fat distribution in lean part of meat is called **marbling**

69. Myoglobin is the pigment in meat that gives the **red color** to meat.

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70. When Myoglobin upon refrigeration changes to **brown red**

71. When Myoglobin upon heating changes **grayish brown**

72. Swelling of muscle fibers with fluid causing the meat to become hard is called **Rigor mortis**

73. Oxidation of **glycogen** brings about production of lactic acid.

74. Stiff muscles of the animal start softening after holding for some time. This process is called **aging**.

75. _____ limit the spoilage rate at temperature below the optimal range can inhibit the microbial growth, enzymatic as well as chemical reactions. **Chilling/refrigeration**

76. Storage of fresh meat is done at a refrigeration temperature of **2 to 5°C**

77. **Smoke** is used some times to give a desirable flavor and to improve the keeping quality of meat.

78. The cereal food contains **65 – 75 %** of carbohydrates.

79. Rice contains about **78%** carbohydrates.

80. Which one of the following includes most common cereal food? **Rice Wheat and Maize**

81. **Bran** is outer chaffy coat that covers the kernel.

82. Germ is a small structure at the lower end of the **endosperm**.

83. Endosperm is constitutes about **84-85%** of the kernel.

84. How many types of wheat? **Two**

85. ____ is also called Corn. **Maize**

86. Oryza sativa (Asian rice) Oryza glaberrima (African rice).

87. Highly polished milled rice has very little **thiamin**

88. Moisture content at time of harvest of Oat should be no more than **13%**

89. Grain is soaked in water until moisture content reaches **50%**

90. Maltose utilized by yeast to produce? **Ethyl alcohol Carbon dioxide**

91. Malt used in the brewing industry convert the starches present in barley, rye, rice, corn, or other grains to **maltose**

92. Buckwheat belongs to the botanical family **Polygonaceae**

93. Triticale with high **lysine** content of rye.

94. Spoilage is defined as **‘’undesirable changes in the properties of a product‘’**

95. **Lipolysis** is the breakdown of fats and other lipids by hydrolysis to release Fatty Acids.

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96. Which of the following pigment forming bacteria? **Flavobacterium Micrococcus and Serratia**

97. Beverage can swells due to the activity of **Clostridium Botulinum**.

98. **LAB (Lactic Acid Bacteria)** produces lactic acid which is used as starter culture for the production of yoghurt from Milk.

99. The best pH for the growth of Micro-organisms is **7.**

100. Psychrotrophs optimum temperature (20-30 ∘ C)

Thermophiles optimum temperature (30-40 ∘ C)

Mesophiles optimum temperature is (20-30 ∘ C)

1 to 50 Lectures

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